

Redlined and Left Behind: Housing Unaffordability and Displacement Risk in Charlotte, NC

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DTSC 2302 Project 3

House Unaffordability & Displacement Risk

Motive/Topic



Key Findings



Regression Model



Classification Model



HOLC Equity Analysis



Conclusion

House Unaffordability & Displacement Risk



Topic Discussion

1

Rising Housing
Costs in
Charlotte /
Difficulty of
Homeownership

2

Concern about
displacement in
vulnerable
neighborhoods

3

"What factors
drive housing
burden and
displacement
risk?"

4

Impact of
historical
redlining

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House Unaffordability & Displacement Risk Key Findings				
GAM R ² is about		XGBoost recall is about	Average precision is about	
0.83		0.83	0.88	
The key findings showed strong link between socioeconomic factors and outcomes. The patterns linked to historical inequality were still visible modern-day.				



Regression Model Overview

The goal of the regression model is to predict the housing cost burden variable.

The models trained and built when reaching the optimal model were:

- Multiple Linear Regression Model (OLS)
- Polynomial Regression Model
- Generalized Additive Model (GAM)

The dataset for this project used contained of:

- 458 observations
- 23 features

The best model achieved was the 4-feature Generalized Additive Model (GAM) model focusing on non-linear relationships.



Regression Model Results (GAM)

The Generalized Additive Model showed key predictors of:

- Household Income
- Homeownership
- Food/nutrition Access
- Age of Residents

Approximate R^2 :

0.83

Approximate RMSE:

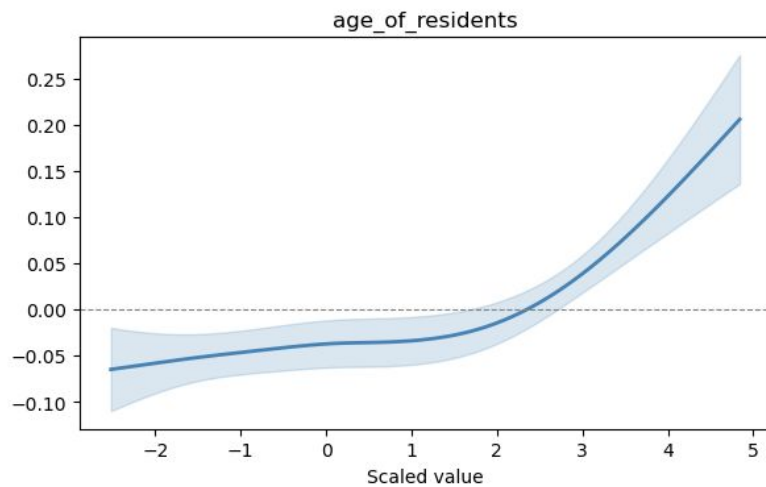
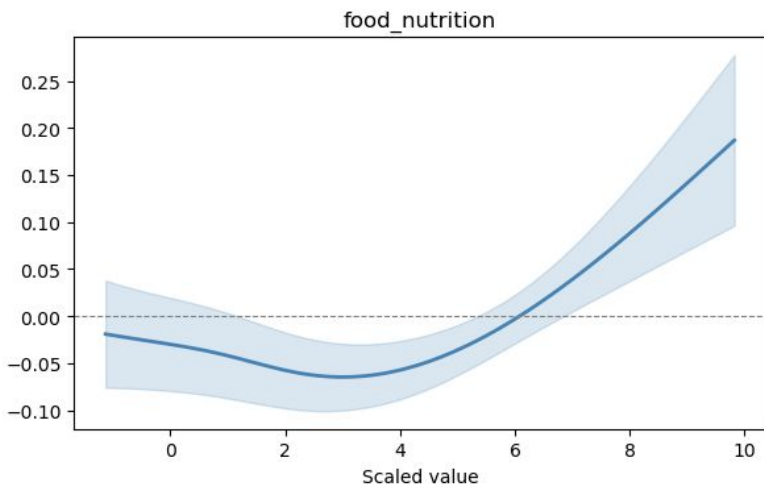
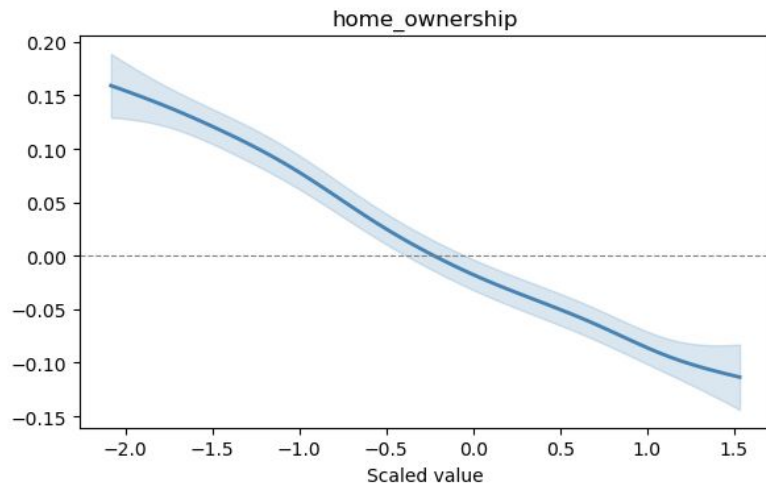
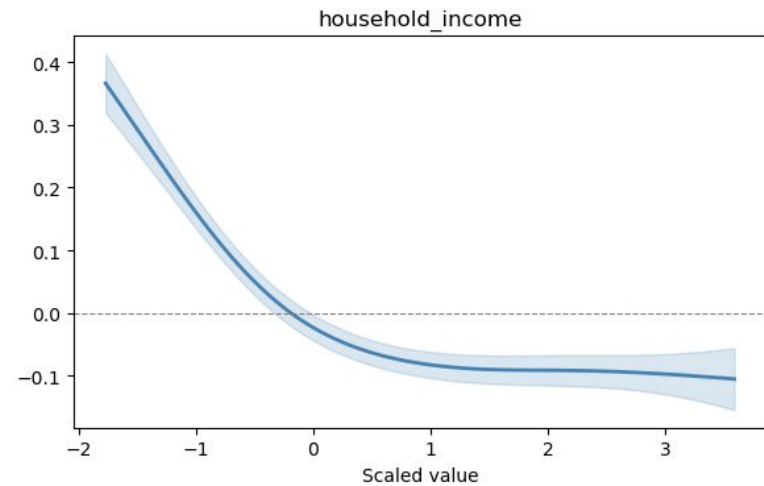
0.066

Cross Validation
Approximate Average R^2 :

0.81

The strong average R^2 shown through the cross validation shows that the model generalizes well to new and unseen data. GAM did not only improve accuracy but also provided more reliable and consistent performance.

GAM (4 Features): Partial Dependence Plots: Housing Cost Burden





Classification Model Overview

The goal of the classification model is to maximize recall for high displacement risk neighborhoods.

The models trained and built when reaching the optimal model were:

- Logistic Regression Model
- Random Forest Model
- K-Nearest Neighbors (KNN)
- XGBoost

The dataset for this project used contained of:

- 458 observations
- 23 features

The best model achieved was the XGBoost model with balance in recall and precision.



Classification Model Results (XGBoost)

Top Predictors of Displacement Risk:

- Household Income
- Food/nutrition Access
- Housing Age
- Code Violations

Approximate Recall:

0.83

Approximate F1:

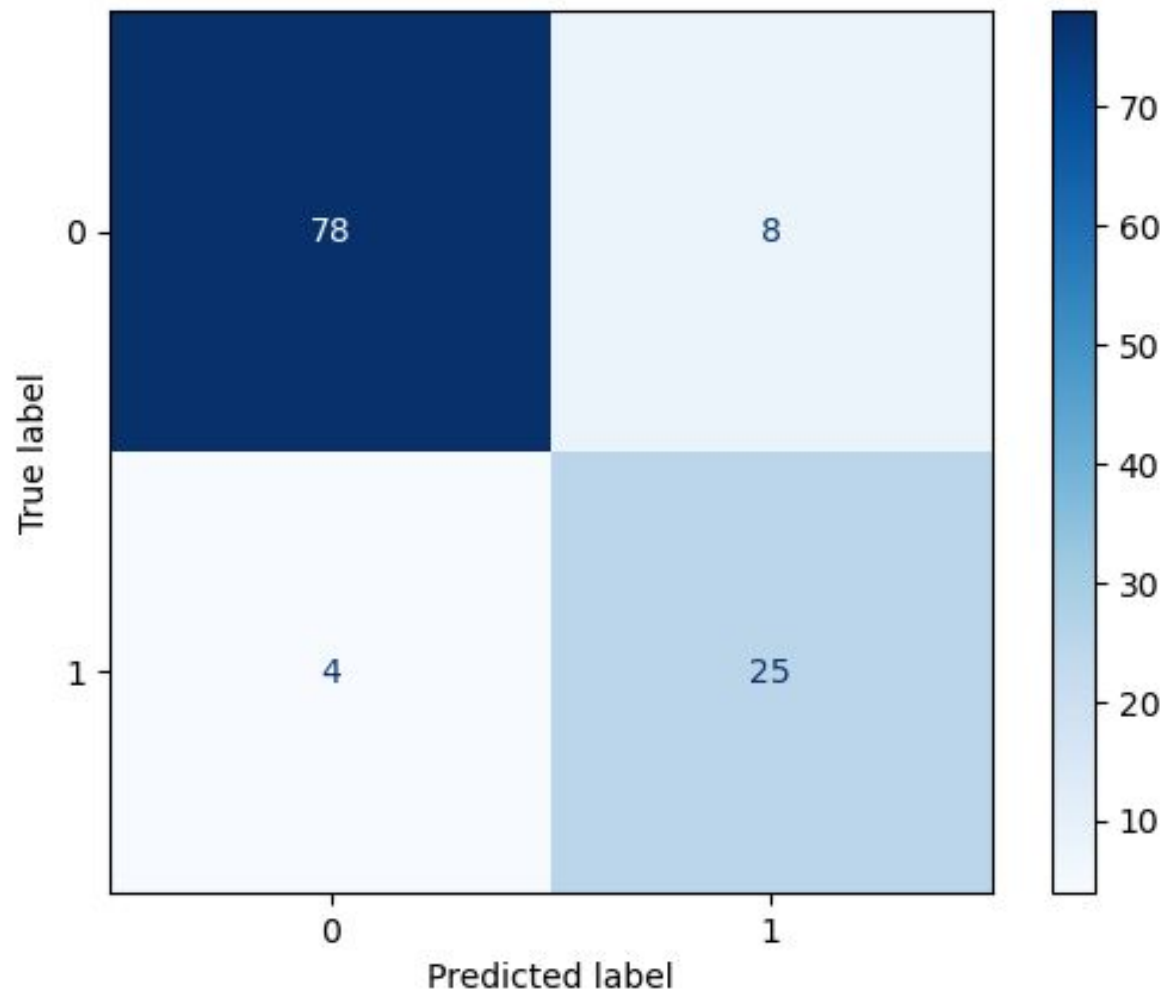
0.79

Approximate Average
Percision:

0.88

This model was chosen as the optimal model due to it having the best balance of percision and recall. It also has the ability to capture nonlinear relationships.

XGBoost Confusion Matrix



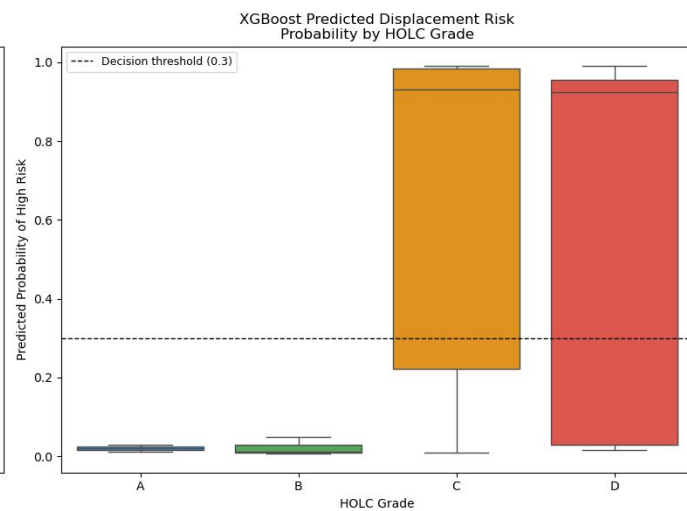
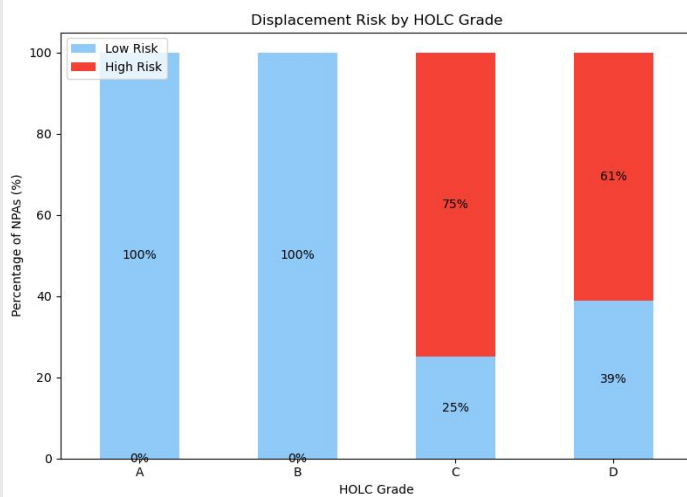
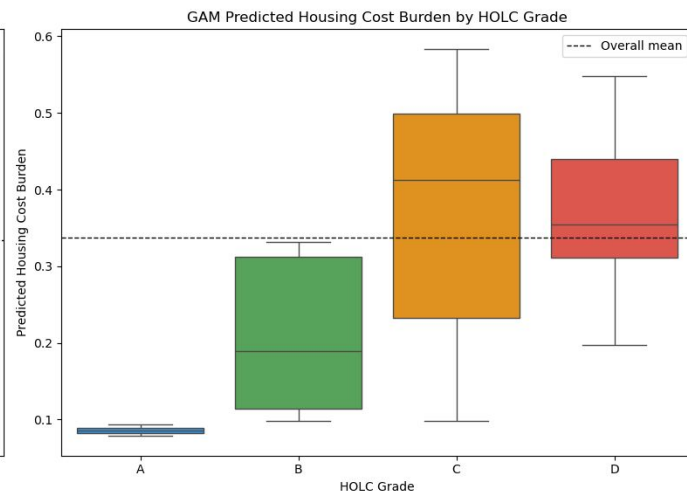
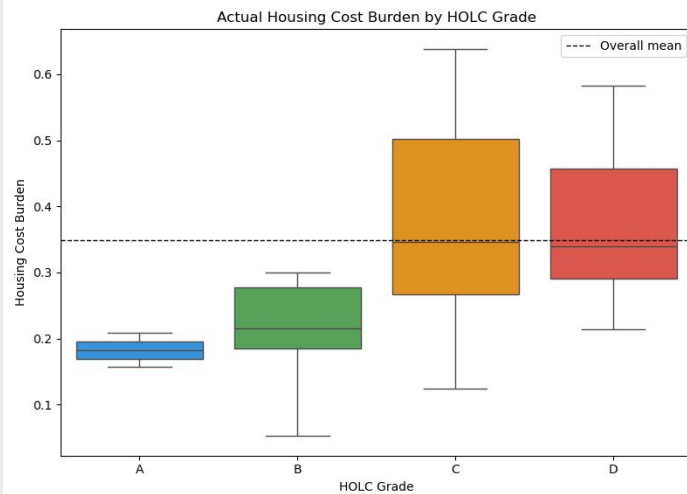
House Unaffordability & Displacement Risk

HOLC Equity Analysis

We also took time to examine whether historical redlining still had any impact on present-day housing outcomes. We found that neighborhoods historically graded C and D had significantly higher housing burden and displacement risk compared to neighborhoods graded A and B.

P-value was shown to be less than 0.05 → statistically significant

A p-value less than 0.05 means the differences in neighborhood grades are unlikely to be due to random, therefore we can treat it to be meaningful. Most importantly HOLC grade was not used in our models, however, the patterns still appeared. The patterns suggested that the historical inequalities due to redlining are still reflected in current-day socioeconomic conditions.





Conclusion

We learned that housing burden is strongly tied to income, home ownership, and neighborhood characteristics.

Best performing models:

- Generalized Additive Model (GAM)
- XGboost Model

These models provided strong predictive performance to help identify high-risk areas.

We found evidence that historical redlining mapping that was initiated in order to mark hazardous neighborhoods in terms of investment still shows impact on current-day neighborhoods.

However, our analysis is still limited by cross-sectional data and does not establish causation. Only association can be established.